FW 0.8.5-7_2026-10-28T18:06:06Z | test 2026-10-30 01:01:34 to 01:24:18 UTC | 400 sps | logger 99977

Folder: 2026-10-29_Lab_0857_TCXO_AdcIntClocks
Pulses analyzed: ADC1 1355, ADC2 1356, ADC3 1356, ADC4 1356

Timing (negative offset = recorded edge later than reference; RMS resid = raw/after per-pulse shift)

ADC	chan	n	offset med (std) ms	theory ms	obs-theory	best-fit shift med (std) ms	RMS resid %
ADC1	HDF	1355	-0.299 (0.013)	-0.000	-0.299	-0.211 (0.029)	1.02/ 0.62
ADC2	EHZ	1356	-7.014 (0.714)	-5.994	-1.019	-1.038 (0.713)	2.57/ 0.31
ADC3	EHN	1356	-6.827 (0.711)	-5.994	-0.832	-0.845 (0.710)	2.25/ 0.30
ADC4	EHE	1356	-6.659 (0.710)	-5.994	-0.664	-0.662 (0.709)	2.01/ 0.35

ADC clocks relative to ADC1 (from the sawtooth of the ADC2-4 offsets)

ADC	chan	clock - ADC1, ppm: median	5-95% over run	ADC clock vs TCXO, ppm
ADC2	EHZ	-605 (alias limit ±1250)	-660 ... -577	+761
ADC3	EHN	+274 (alias limit ±1250)	+243 ... +305	+1640
ADC4	EHE	+99 (alias limit ±1250)	+59 ... +120	+1465

ADC1 clock vs TCXO (log adc_ppm median): +1366 ppm. ADC1 offset trend: +64.3 us/hour.

How to read this report

- Theory = the signal that should have been recorded: ideal 1PPS step through the analog filter (ADC2-4 only) and the ADC sinc3 filter, sampled on the UTC grid, then passed through the same analysis as the data (levels, padding, aliasing).
- sinc3 is zero-phase: firmware time-tags each sample at the center of the sinc3 window.
- ADC2-4 analog model: ladder, corner 48.25 Hz per stage, stage-2 impedance 10 x stage 1.
- Edge plots: time axis is ((t + 0.5) mod 1) - 0.5 seconds from the UTC second. Spectra: DFT of each pulse relative to an ideal step (periodic square-wave padding), so only odd harmonics of 1 Hz exist; negative frequencies mirror positive ones; phase is unwrapped outward from low frequency. Envelope = mean ± 2 std over ALL pulses (<=150 individual drawn).
- Offset = phase slope through the origin below 40 Hz. Best-fit shift = time shift of the theoretical edge that minimizes squared error against each observed edge in the time domain.
- ADC2-4 are read at ADC1's DRDY while converting on their own clocks, so each sample is 0..T older than ADC1's (T = sample period): expected mean extra lag T/2 and offset scatter T/sqrt(12) = 0.722 ms (observed std: ADC2 0.714, ADC3 0.711, ADC4 0.710 ms).
- Clock beat: the offset sawtooth advances (clock difference x 1 s) per pulse, wrapped to +/- T/2. Sign: a faster ADC clock gives older data, so clock difference (ppm) = -(offset increment, ms/s) x 1000. Valid within the alias limit.

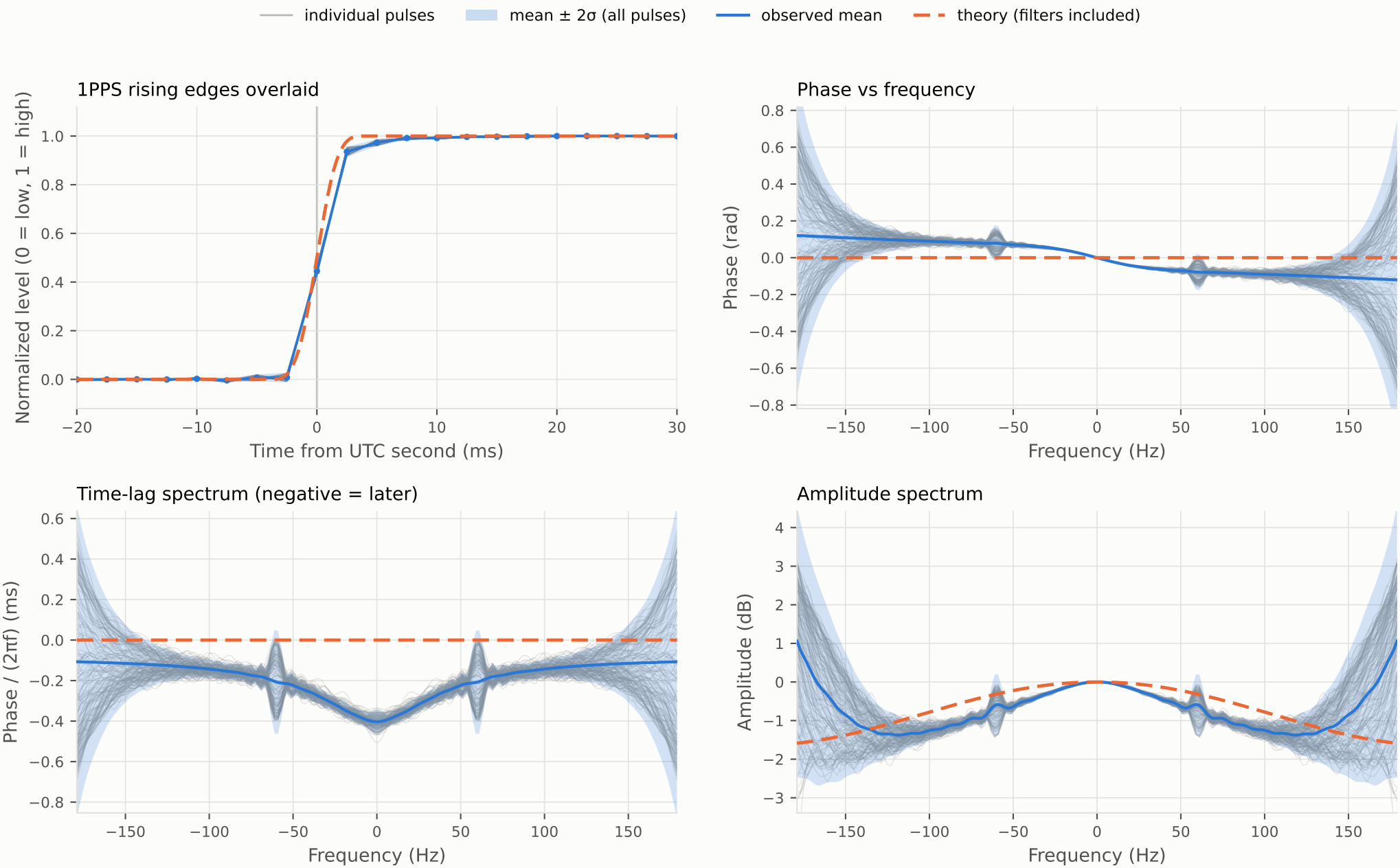
ADC1 (HDF): no analog filter + sinc3

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Time offset (negative = later), phase slope 0-40 Hz: observed median -0.299 ms (std 0.013) theory -0.000 ms observed - theory -0.299 ms

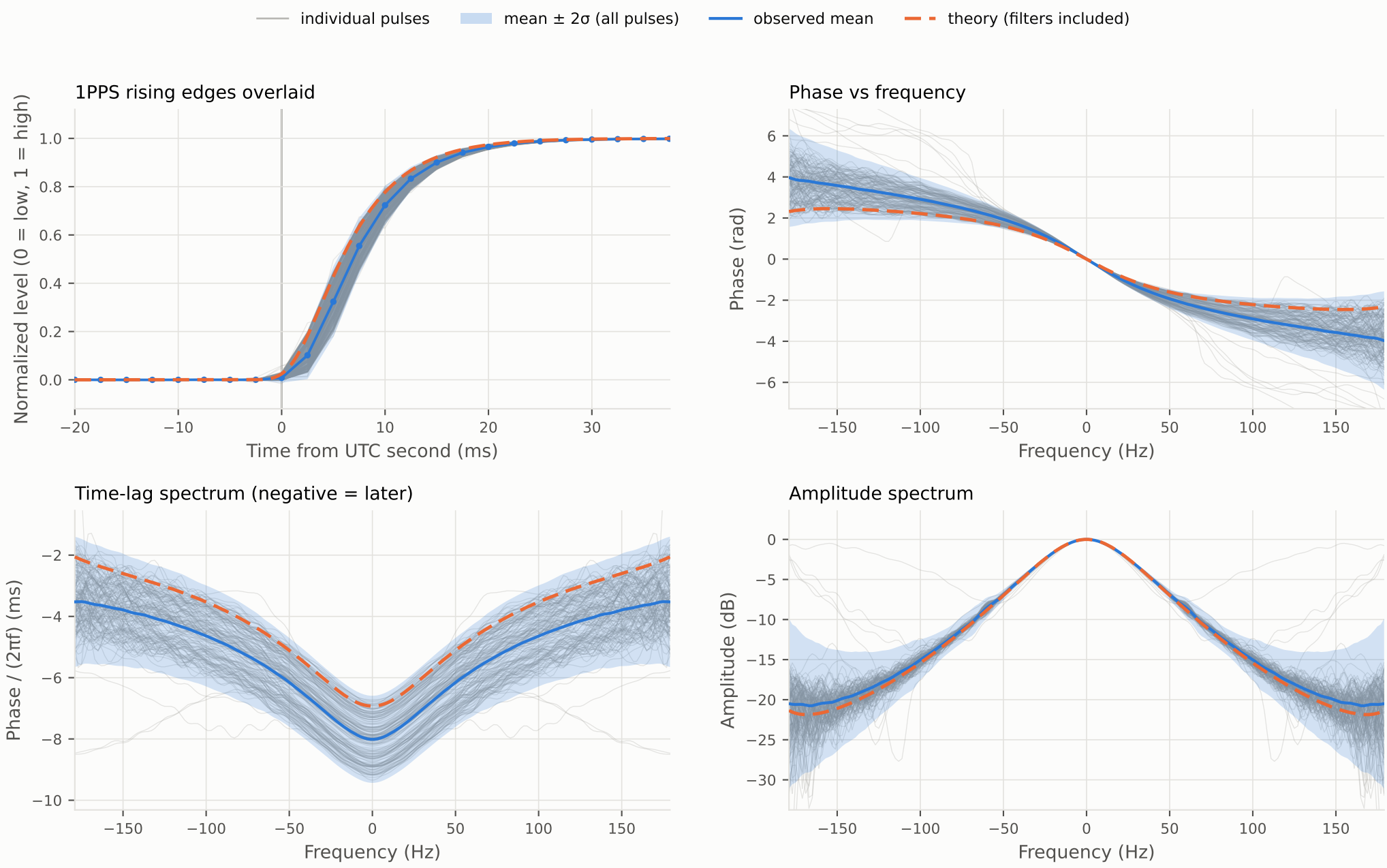
Per-pulse best-fit shift of theoretical edge: median -0.211 mean -0.212 std 0.029 2.5-97.5%: [-0.258, -0.161] ms

Edge residual RMS (% of step): 1.02 raw, 0.62 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.086 rad, amplitude 0.73 dB



ADC2 (EHZ): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.5-7_2026-10-28T18:06:06Z | test 2026-10-30 01:01:34 to 01:24:18 UTC | 400 sps | logger 99977
Time offset (negative = later), phase slope 0-40 Hz: observed median -7.014 ms (std 0.714)   theory -5.994 ms   observed - theory -1.019 ms
Per-pulse best-fit shift of theoretical edge: median -1.038 mean -1.053 std 0.713 2.5-97.5%: [-2.230, +0.144] ms
Edge residual RMS (% of step): 2.57 raw, 0.31 after per-pulse shift.   Mean spectrum vs theory, RMS: phase 0.786 rad, amplitude 0.59 dB



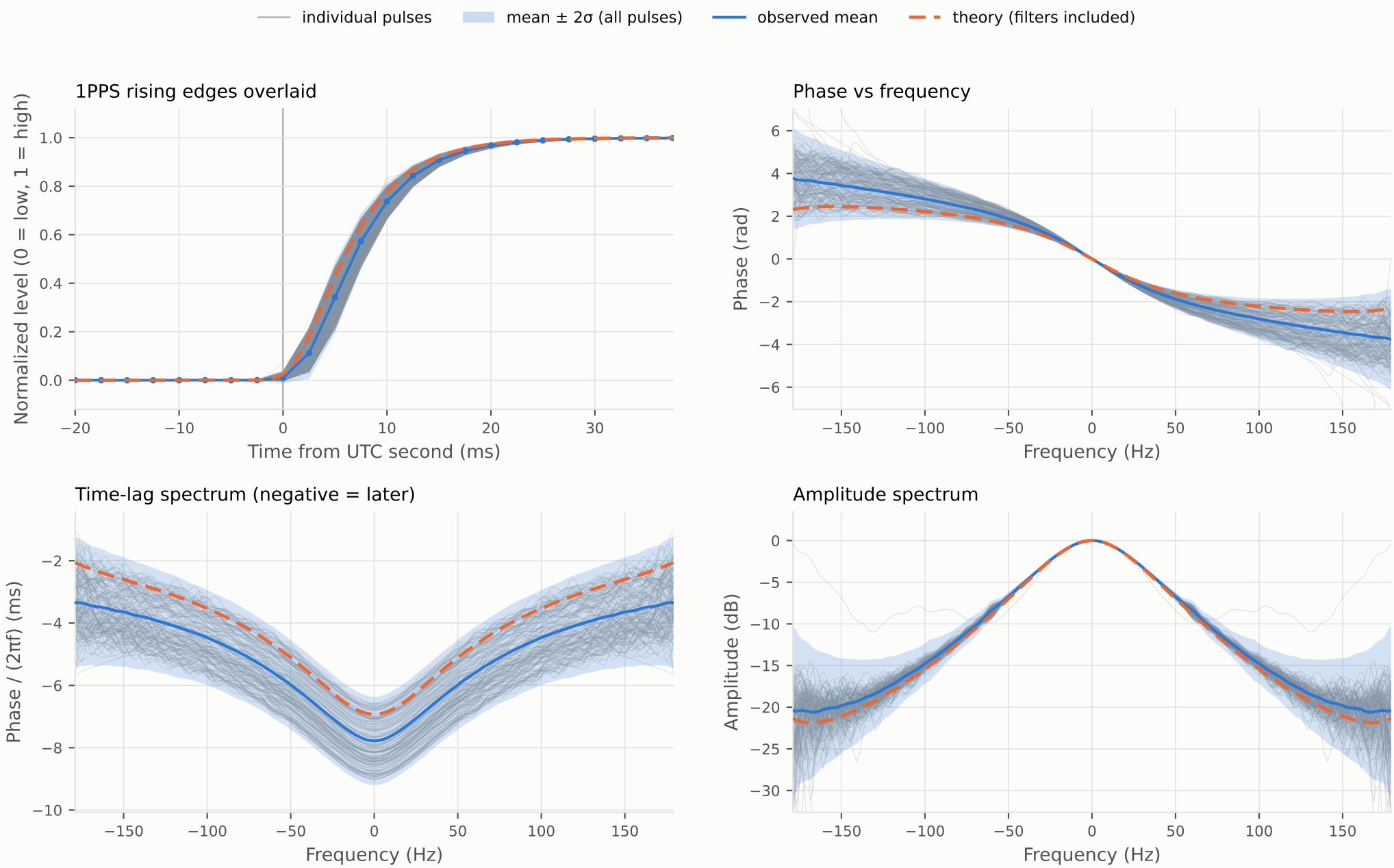
ADC3 (EHN): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.5-7_2026-10-28T18:06:06Z | test 2026-10-30 01:01:34 to 01:24:18 UTC | 400 sps | logger 99977

Time offset (negative = later), phase slope 0-40 Hz: observed median -6.827 ms (std 0.711) theory -5.994 ms observed - theory -0.832 ms

Per-pulse best-fit shift of theoretical edge: median -0.845 mean -0.848 std 0.710 2.5-97.5%: [-2.030, +0.322] ms

Edge residual RMS (% of step): 2.25 raw, 0.30 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.686 rad, amplitude 0.75 dB



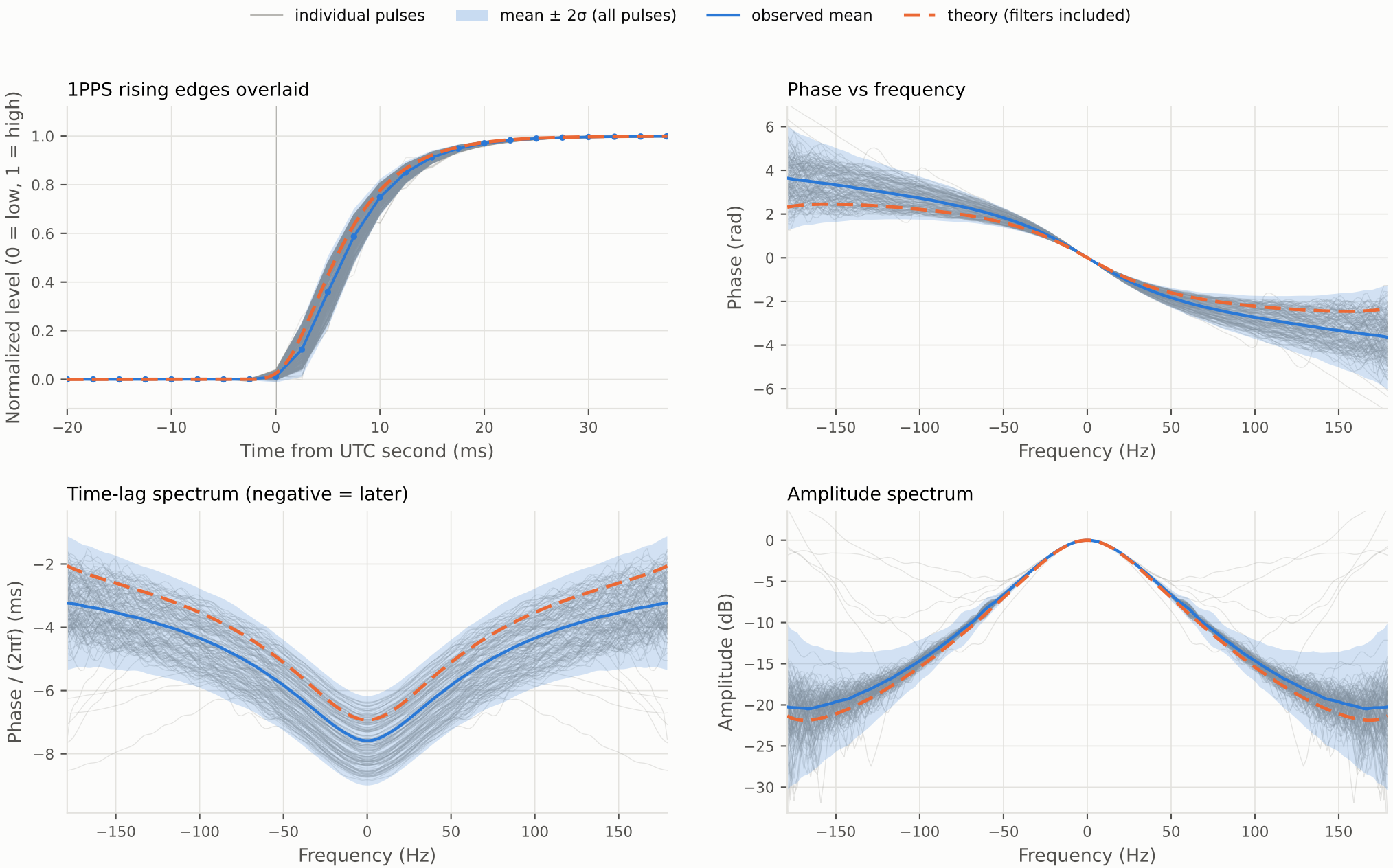
ADC4 (EHE): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.5-7_2026-10-28T18:06:06Z | test 2026-10-30 01:01:34 to 01:24:18 UTC | 400 sps | logger 99977

Time offset (negative = later), phase slope 0-40 Hz: observed median -6.659 ms (std 0.710) theory -5.994 ms observed - theory -0.664 ms

Per-pulse best-fit shift of theoretical edge: median -0.662 mean -0.673 std 0.709 2.5-97.5%: [-1.860, +0.494] ms

Edge residual RMS (% of step): 2.01 raw, 0.35 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.608 rad, amplitude 0.87 dB



Time offset and ADC clock beat vs time

FW 0.8.5-7_2026-10-28T18:06:06Z | test 2026-10-30 01:01:34 to 01:24:18 UTC | 400 sps | logger 99977

Offsets from the phase slope below 40 Hz (negative = recorded edge later); dashed gray = theory.

